Pinned group summary: `SU(n=2, q=1, eps=1)`

Pinning-Sympy automated output

July 6, 2026

Contents

1 Basic group attributes

Group	`SU(n=2, q=1, eps=1)`
Matrix size	2×2
Rank	1
Defining equations	$DefiningEquationPlaceholder$

1.1 Hermitian form

Dimension	2
Witt index	1
Primitive element	$\sqrt{d}$
Epsilon	1

Matrix: $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

1.2 Generic Lie algebra element

$$\begin{bmatrix} x_{r00} & \sqrt{d}x_{i01} \\ \sqrt{d}x_{i10} & -x_{r00} \end{bmatrix}$$

1.3 Generic torus element

$$\begin{bmatrix} t_0 & 0 \\ 0 & \frac{1}{t_0} \end{bmatrix}$$

1.4 Trivial characters

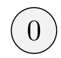
$$\begin{bmatrix} 1 & 1 \end{bmatrix}$$

2 Root system

Summary Properties

Dynkin type	`A1`
Irreducible	True
Reduced	True
Simply laced	True
Number of roots	2

2.1 Dynkin diagram



2.2 Cartan matrix

$$\begin{bmatrix} 2 \end{bmatrix}$$

2.3 Roots and coroots

Root	**Coroot**	**Simple**	**Sign**	**Multipliable**
$(-2,\ 0)$	$(-1,\ 1)$	No	$-$	No
$(2,\ 0)$	$(1,\ -1)$	Yes	$+$	No

2.4 Linear combinations of roots

Definition 2.1. For two roots α, β in a root system Φ, the set of positive integral linear combinations of α, β that are also roots is denoted $(\alpha, \beta) = \{i\alpha + j\beta \in \Phi : i, j \in \mathbb{Z}_{\geq 1}\}$

No pairs of roots generate positive integer linear combinations.

3 Root spaces

Root	Dimension
$(-2,\ 0)$	1
$(2,\ 0)$	1

RootSpaceTablePlaceholder

4 Root subgroups

RootSubgroupTablePlaceholder
HomomorphismDefectTablePlaceholder
HomomorphismDefectIdentitiesPlaceholder

5 Commutators

CommutatorCoefficientTablePlaceholder
CommutatorIdentitiesPlaceholder

6 Weyl group

WeylGroupPlaceholder
WeylConjugationCoefficientPlaceholder
WeylConjugationIdentitiesPlaceholder

7 Coroot torus elements (h_α)

CorootTorusElementPlaceholder